

Mail Defender Case Studies

Suspicious Email Analysis and Phishing Detection System
College Submission Document | Prepared in June 2026

Project Overview

Mail Defender is a web-based email threat analysis platform. It helps users inspect suspicious emails before clicking links, opening attachments, replying to the sender, or trusting the message. The system accepts pasted headers/body or uploaded email files, extracts important evidence, checks sender authentication, enriches indicators with threat intelligence, calculates a risk score, and generates a report.

Purpose of This Case Study Document

- To demonstrate how Mail Defender can be applied to realistic suspicious email situations.
- To show how evidence such as SPF, DKIM, DMARC, URLs, domains, IP addresses, reply-to mismatch, and attachment hashes is used during analysis.
- To explain the output expected from the system: verdict, risk score, key findings, attack type, and recommended action.
- To connect the current project with future company-level upgrades such as saved cases, dashboards, audit logs, SIEM export, and team workflows.

Mail Defender Module	Role in Case Study Analysis
Email Intake	Receives pasted raw headers/body or uploaded .eml/.txt files from the user.
Parser and Header Review	Separates sender details, subject, routing information, authentication results, and body content.
IOC Extraction	Finds URLs, domains, IP addresses, email addresses, hashes, and attachment evidence.
Threat Intelligence	Checks indicators through VirusTotal and IP reputation through AbuseIPDB when API keys are available.
Risk Scoring and Report	Calculates a 0-100 score, assigns a verdict, summarizes evidence, and supports PDF export.

Case Study 1: Credential Phishing Email

Scenario: A student or employee receives an urgent message claiming that their college or company mailbox will be disabled within 24 hours. The email asks the user to verify the account by clicking a login link. The display name looks official, but the sender domain and reply-to address are different from the trusted organization domain.

Aspect	Details
Attack Goal	Steal login credentials by sending the user to a fake sign-in page.
User Risk	Account takeover, mailbox misuse, password reuse attack, and possible data exposure.
Input to Mail Defender	Raw headers, subject, body text, and suspicious verification link.
Expected Verdict	High Risk or Suspicious, depending on reputation and authentication results.

Evidence Detected by Mail Defender

- Authentication-Results header may show SPF fail, DKIM fail, or DMARC issue, which weakens sender trust.
- Reply-To address is different from the visible From address, suggesting impersonation or redirection.
- Body contains urgency keywords such as verify now, account blocked, final warning, or login required.
- IOC extractor identifies the external login URL, root domain, and any IP addresses present in links or headers.
- VirusTotal can be used to check whether the link or domain has known malicious or suspicious detections.

System Processing Flow

- The user submits the suspicious email through the Mail Defender workspace.
- The backend parses headers and message body, then extracts links, domains, emails, and IPs.
- The authentication panel shows SPF, DKIM, DMARC, reply-to mismatch, and spam status.
- The scoring engine increases risk because of credential language, authentication failure, and suspicious URL evidence.
- The report section generates a readable summary with key findings and safe handling steps.

Recommended Action

Do not click the link. Verify the account warning through the official college/company portal. Report the email to IT or security. If credentials were entered, reset password immediately and review account activity.

Case Study 2: Business Email Compromise / Payment Fraud

Scenario: A finance or administration user receives an email that appears to come from a supplier, manager, or college vendor. The message requests urgent payment approval, bank account change, or invoice settlement. The email may not contain a known malicious link, but the wording, sender identity, and reply-to pattern are suspicious.

Aspect	Details
Attack Goal	Trick the organization into transferring money or changing payment information.
User Risk	Financial loss, vendor dispute, approval fraud, and reputational damage.
Input to Mail Defender	Headers, sender details, subject line, body message, and any invoice link or attachment.
Expected Verdict	Suspicious or High Risk because business context and identity mismatch matter even without malware.

Important Observations

- The display name may look trusted, but the actual email address can be from a free mailbox or look-alike domain.
- The Reply-To value may route the conversation to the attacker instead of the real vendor or manager.
- The body may use pressure tactics such as urgent, confidential, approve today, or do not call.
- If an invoice link exists, Mail Defender extracts the URL/domain and checks it through reputation services.
- Even when external reputation is clean, local risk scoring can still mark the email suspicious because BEC attacks often use newly created or low-volume infrastructure.

How Mail Defender Helps

- It presents header details and authentication results in one place so the user can compare visible sender and technical sender.
- It highlights reply-to mismatch, suspicious language, and financial lure terms.
- It generates a report that can be shared with a manager, IT helpdesk, or finance approver before any payment action is taken.
- It supports evidence-based decision making instead of relying only on the email appearance.

Recommended Action

Do not approve payment from email alone. Confirm bank account or invoice changes through a trusted phone number or official vendor portal. Preserve the email and export the Mail Defender report for review.

Case Study 3: Malware Attachment / Fake Invoice

Scenario: A user receives an email with an attachment named like invoice.pdf, payment_notice.zip, document.html, or statement.docx. The message asks the user to open the file to view billing details. This type of attack attempts to deliver malware, steal credentials through an HTML page, or lure the user into downloading another payload.

Aspect	Details
Attack Goal	Deliver malicious file, steal credentials, or lead the victim to a malware download page.
User Risk	Endpoint infection, credential theft, data loss, and lateral movement in the network.
Input to Mail Defender	Uploaded .eml file containing attachment metadata and message content.
Expected Verdict	High Risk when attachment hash, link reputation, authentication failure, or malware language is detected.

Evidence Detected by Mail Defender

- Attachment details are extracted from the email file, including filename, content type, size, and SHA-256 hash.
- The computed hash is added to the IOC list so it can be checked through VirusTotal.
- The body is scanned for attachment-related lures such as invoice attached, download document, macro, or payment notice.
- Any embedded links are extracted and checked as URL/domain indicators.
- Authentication failures and suspicious sender mismatch increase the overall risk score.

System Outcome

- The dashboard shows the attachment under Attachment Review with its SHA-256 value.
- The IOC board lists URLs, domains, IP addresses, and hashes grouped by type.
- The risk report explains why opening the attachment is unsafe and what indicators were observed.
- The exported PDF can be attached to an incident ticket or used for academic submission evidence.

Recommended Action

Do not open the attachment on a normal system. Isolate the email, check the file hash, block malicious indicators, and escalate to IT/security for sandbox or endpoint investigation if required.

Case Study 4: Safe Email Verification / False Positive Control

Scenario: A user submits a legitimate college notice, HR update, or internal information email to Mail Defender because they are unsure whether it is safe. The sender domain is correct, authentication checks pass, links point to official domains, and the message does not contain pressure tactics or suspicious attachments.

Aspect	Details
Purpose	Show that Mail Defender is not only for finding attacks; it also helps users verify safe communication.
User Risk	Low, provided the domain and authentication evidence are consistent.
Input to Mail Defender	Normal email headers/body or uploaded .eml/.txt file.
Expected Verdict	Safe or Low Risk when evidence is clean and no suspicious IOC is found.

Clean Signals

- SPF, DKIM, and DMARC results pass or align with the trusted sender domain.
- Reply-To address matches the expected organization or sender identity.
- Links point to official domains instead of shortened URLs or look-alike domains.
- No suspicious attachment hash, malicious URL reputation, or abuse-heavy IP reputation is detected.
- The generated report explains why the message appears low risk, while still advising careful review before taking sensitive action.

Comparison of Case Study Results

Case	Main Threat	Key Evidence	Likely Verdict
1	Credential phishing	Fake login URL, urgency language, authentication issue, reply-to mismatch	High Risk
2	BEC/payment fraud	Display-name spoofing, financial lure, reply-to mismatch, unusual request	Suspicious/High Risk
3	Malware attachment	Attachment hash, suspicious file lure, link/domain indicators	High Risk
4	Legitimate notice	Authentication passes, official domain, no malicious IOC	Safe/Low Risk

Learning Outcomes and Company-Level Upgrade Scope

The case studies show that Mail Defender supports both technical and practical email investigation. It does not depend on only one signal. Instead, it combines sender authentication, content analysis, IOC extraction, threat intelligence, local scoring, and reporting to create a complete decision-making view for the user.

Key Learning Outcomes

- A phishing email can look professional, so header evidence and sender authentication are important.
- Business email compromise can be dangerous even when no malware is attached and reputation tools show limited results.
- Attachment hashing is useful because the same file can be checked, blocked, or shared with security teams using a stable SHA-256 value.
- A safe verdict is also valuable because it reduces confusion and teaches users how legitimate emails usually appear.
- A structured PDF report makes the analysis useful for college submission, helpdesk escalation, or incident documentation.

Future Company-Level Enhancements Based on These Cases

Upgrade Area	How It Improves Case Handling
Saved Case Management	Store each suspicious email investigation with status, owner, timestamps, exported report, and evidence history.
User Accounts and Roles	Allow students, helpdesk users, analysts, and admins to access different features securely.
Organization Dashboard	Show top attack types, high-risk emails, common malicious domains, and analysis volume trends.
SIEM/Ticketing Export	Send high-risk findings to tools such as Jira, ServiceNow, or a SIEM for faster response.
Attachment Sandbox	Analyze risky files in an isolated environment instead of only checking hashes.
Audit Logs and Retention	Track who analyzed, exported, updated, or closed a case for accountability.

Final Note for Submission

These case studies are designed as academic examples for Mail Defender. They describe realistic usage, expected evidence, system behavior, and recommended response actions without requiring real malicious samples.